

Sara Norouzinia

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Education

Georgia Institute of Technology	Atlanta, GA
<i>Bachelor of Science in Computer Science (GPA: 3.74)</i>	May 2027
<i>Master of Science in Computer Science, Specialization in Machine Learning</i>	May 2028
• Relevant Coursework: Object-Oriented Programming; Data Structures & Algorithms; Introduction to AI; Computer Organization and Programming; Systems and Networks; Design and Analysis of Algorithms; Game AI	

Technical Skills

Languages: Python, Java, C/C++, Assembly, JavaScript, TypeScript, HTML, CSS, Bash
Frameworks & Tools: React, Node.js, Django, FastAPI, Docker, Jest UI, PyTorch, MongoDB, Jira, VSCode, Overleaf, Bootstrap, Git, GitHub, Figma, Linux/Unix
Development Practices: Agile, Scrum, CI/CD, Web Dev, System Design & Architecture, Machine Learning, NLP, CNN

Experience

• Software Engineering Intern	May 2026 – August 2026
<i>GTRI Applied Systems Lab</i>	Huntsville, AL
• Wrote 6 end-to-end automation test suites in Python and C++ to validate MongoDB request handling, surfacing read/write operation bugs before they reach production	
• Resolved XML defects in a middleware layer between MongoDB to a web app, improving cross-service communication reliability	
• Refactored 5 React components for web application responsiveness and accessibility, extending Jest UI for testing	
• Software Engineering Intern	May 2026 – August 2026
<i>TechX AI Systems and LLM Engineering</i>	
• Built a filter-based agent-lookup feature across 5 FastAPI endpoints, supporting sort/filter by popularity, relevance, & metadata	
• Authored 8 backend API tests covering 4 endpoints and developed prompt templates, mock runners, and tracing instrumentation to streamline debugging	
• Implemented tool-selection and routing logic for an agentic AI system using MCP	
• Machine Learning Researcher	Jan 2026 – May 2026
<i>Anomaly Sound Detection in the Ocean Soundscape</i>	Metz, France
• Accepted for International Conference on Pattern Recognition and published in Springer's LNCS series	
• Programmed and trained a supervised ConvNeXt-based CNN in PyTorch for binary anomaly classification on a 600 audio dataset and validated against a 100-sample expert-labeled ground truth	
• Built a deep learning pipeline tuning via BCEWithLogitsLoss, early stopping, and ReduceLROnPlateau scheduling with L2-regularized Adam optimization to stabilize convergence on high-variance acoustic data	
• Evaluated model performance using F1-score, precision/recall metrics, and confusion matrix	
• Machine Learning Engineer	Aug 2025 – Present
<i>Data Driven Education – VIP Undergraduate Research</i>	Atlanta, GA
• Collaborated in a cross-functional team to fine-tune a BART model for automated question-difficulty classification and generation on real student exam data	
• Built a linguistic feature-extraction pipeline using NumPy and textstat, computing readability metrics to score difficulty	
• Architected a dual-pipeline system (generation & classification) to cross-validate model outputs and improve prediction reliability	

Projects & Research

• Align Web App – Backend Developer	Jan 2025 – May 2025
• Built backend infrastructure using Django models and Google OAuth2, integrating Google Calendar to support 10+ app features	
• Engineered event-driven triggers to automate system actions based on real-time user state changes, reducing manual coordination	
• Collaborated in a 5-person Agile team, coordinating schema and API design through GitHub version control and sprint planning	
• Personal Portfolio Website	Jul 2025
• Modeled and produced a fully responsive personal website using React, TypeScript, JavaScript, HTML, and CSS	
• Implemented reusable components to create a scalable, maintainable front-end application	
• Prototyped end-to-end UI/UX flows and wireframes in Figma and Milanote prior to implementation	
• Vehicle Gas Consumption Analysis using IoT and GIS	Aug 2023 – May 2024
• Designed a data-driven system integrating IoT data and GIS mapping to analyze vehicle fuel consumption patterns	
• Conducted a one-group pre-test/post-test experimental study across 10 random participants	
• Built data pipelines to process, validate, and analyze large-scale sensor datasets for statistical analysis	

Leadership

• GT Persian-American Resource Society – Executive Board	
Planned 10+ events, facilitated member engagement, & coordinated with UGA and KSU clubs	Atlanta, GA
• Teaching Assistant – Object-Oriented Programming	
Led 20+ student review sessions, supported individuals in office hours, and graded deliverables	Atlanta, GA
• 1st Georgia Tech IBM SkillsBuild Student Ambassador	
Connected with future IBM skillbuilders and taught them about the importance of AI skills	Atlanta, GA
• Team Lead – Kaldi's Coffee	
Ran daily operations across 3 cafes, supported staff, and delegated team responsibilities	Atlanta, GA